

Python Task

- 1) Print the following exactly (including quotes): "Python\nRocks".
- 2) Swap the values of a and b without using a third variable.
- 3) Compute $3^{**}2 + 10//3 - 7\%3$.
- 4) Given $x=7$, update x using an in-place operator to add 5.
- 5) Convert string "42.7" to an integer safely (round down).
- 6) Identify the types of: 42, 3.14, True, "hi", [1,2], (1,2), {1,2}, {"a":1}.
- 7) Read two integers from input and print their sum.
- 8) Given $s="python"$, produce "PYThon" using slicing and methods.
- 9) Given $s="abcdef"$, get every second character starting from index 1.
- 10) Grade system: input score 0–100, print A (≥ 90), B (80–89), C (70–79), D (60–69), F (< 60).
- 11) Count occurrences of "ana" in "bananana" (overlaps allowed? Use .count for non-overlap.)
- 12) Compute the sum of a list [1,2,3,4] without using sum().
- 13) Read integers until the user enters 0, then print the total.
- 14) Loop 1..10 and stop at the first number divisible by 7; print that number only.
- 15) Write a function area_circle(r) returning area (πr^2). Use math.pi.
- 16) Write a function with a default argument: greet(name, msg="Hello") that returns "Hello, Alice" when called as greet("Alice").
- 17) Create a list of squares for numbers 1..5 using a list comprehension.
- 18) Flatten [[1,2],[3,4]] into [1,2,3,4] (simple one-level).
- 19) Remove the first occurrence of 1 from nums and then reverse the list.
- 20) Given a string input of numbers separated by commas (e.g., "10,20,30"), compute and print their average to 2 decimal places.